AI Resume & Job Matching System

Powered by Endee Vector Database + Google Gemini RAG

Semantic Search | Vector DB | RAG | Flask | Gemini AI

? Problem Statement

Keyword Mismatch

Traditional portals miss candidates who use different words for same skills (ML vs Machine Learning)

Manual Screening

HR teams spend hours reading hundreds of resumes manually for every job opening

Poor Matching

Simple filters cannot understand context — a Python dev who knows SQL gets wrongly filtered out

Lost Talent

Great candidates get rejected because resume lacks exact keyword matches required by system

Our Solution

AI-powered semantic matching that understands the MEANING of skills and experience

Semantic Search

Understands meaning using AI embeddings — beyond keywords

Endee Vector DB

Stores 384-dim vectors with cosine similarity search

Gemini RAG

Generates match explanations and interview questions

PDF Upload

Upload resume PDF — auto extract text and match to jobs

Dual Matching

Match resumes to jobs or job descriptions to candidates

Match Score

Similarity percentage for every result with visual bar

System Architecture

INGESTION PIPELINE

Documents → Chunking → Embeddings → Endee DB

QUERY PIPELINE

Query → Embed → Endee → Results → Gemini → Answer

TECH STACK

Component	Technology	Purpose
Vector Database	Endee (Docker)	Store & search embeddings
Embeddings	all-MiniLM-L6-v2	384-dim semantic vectors
RAG / LLM	Google Gemini 1.5 Flash	Explain matches + interview Qs
Backend	Python Flask	REST API server
Frontend	HTML / CSS / JS	Web UI with 4 tabs

⍰ How Endee Vector Database is Used

01

Create Index

Single index stores both resumes and jobs with INT8 precision

```
client.create_index(
  name='resume_jobs',
  dimension=384,
  space_type='cosine',
  precision=Precision.INT8
)
```

02

Upsert Vectors

Each resume/job stored as 384-dim vector with metadata

```
index.upsert([{
  "id": "resume_001",
  "vector": [...384 floats...],
  "meta": {"type":"resume",...}
}])
```

03

Semantic Query

Top-K similar vectors returned — no keywords needed

```
results = index.query(
  vector=query_embedding,
  top_k=5
)
# Returns cosine-similar items
```

Google Gemini RAG Integration

Endee Search
Top-K chunks retrieved

→

Build Prompt
Resume + Job context

→

Gemini API
gemini-1.5-flash

→

RAG Output
Structured JSON

GEMINI GENERATES:

Match Explanation

2-3 sentences explaining why candidate fits the role

Strengths

Top 3 skills that align with job requirements

Skill Gaps

Missing skills or areas needing development

Interview Questions

5 tailored questions based on the specific match

⍰ Application Features

🔍 Smart Search

Type any skill query — Endee returns semantically matched jobs or candidates instantly with similarity scores

📄 Match My Resume

Paste resume text → system finds best matching jobs from Endee index using vector similarity search

💼 Find Candidates

Post a job description → system finds best matching resumes stored in Endee vector database

📤 Upload PDF

Upload resume PDF → auto extract text → embed → search Endee → show matched jobs with scores

20 Vectors in Endee	10 Resumes	10 Job Descriptions	384 Dimensions	Cosine Similarity

Project Structure & Setup

```
endee-resume-matcher/
  app.py          # Flask server
  search.py       # Endee search
  data.py         # Seed data
  rag.py          # Gemini RAG
  docker-compose.yml
  requirements.txt
  README.md
  templates/
    index.html    # Web UI
```

1. `docker compose up -d`

   Start Endee on port 8080

2. `pip install -r requirements.txt`

   Install Python packages

3. `python data.py`

   Ingest 20 vectors into Endee

4. `python app.py`

   Start Flask on port 5000

5. `http://localhost:5000`

   Open in browser

Sample Results

Query: "Python Data Analyst with Power BI and SQL" → Mode: Find Jobs

#1 **Data Analyst**

Amazon India

Python · SQL · Power BI · Tableau

51% match

#2 **Power BI Developer**

Deloitte India

Power BI · DAX · SQL · ETL

44% match

#3 **Senior Data Scientist**

Flipkart

Python · ML · Statistics · SQL

38% match

✨ Click AI Explain on any result → Gemini generates match explanation + 5 interview questions

Thank You!

AI Resume & Job Matching System — Endee Vector DB + Google Gemini RAG

☆ Starred endee-io/endee Forked endee-io/endee Submitted on GitHub Tap Academy — Endee.io

Built with: Endee · sentence-transformers · Flask · Google Gemini · Docker